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Reversible solutions

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Improved technologies are required to curb the substantial greenhouse gas emissions associated with heating and cooling. Researchers have now revisited endothermic cooling to design a sustainable, reversible sorption-driven dissolution refrigeration cycle, which holds broad potential for cooling, heating, and thermal storage applications.

Heating and cooling account for approximately 40% of global energy-related CO_2 emissions[1]. Vapour compression, the dominant refrigeration and air conditioning technology, consumes roughly 20% of global electricity and contributes 7.5% to global warming, with 30% of that impact resulting from fluorinated refrigerant leakage[2]. To reduce greenhouse gas emissions from this sector, environmentally friendly alternatives are essential. However, the solutions developed so far face considerable challenges.

For example, refrigerants with lower global warming potential often pose toxicity, flammability, and explosion risks, driving research into developing sustainable and more efficient alternatives for refrigeration and heating. Solid-state caloric cooling and heating technologies[3,4] offer a promising alternative to conventional refrigeration by exploiting near-reversible caloric effects in functional solid materials, which are leak-free and potentially recyclable. However, their low power density means that sustained and intensive research efforts are needed. Another alternative technology, electrochemical (redox) refrigeration[5], leverages the thermogalvanic effect to produce cooling by harnessing entropy changes in ions during redox reactions driven by applied electrical work. Despite its promise due to the absence of moving parts, the very low second-law efficiency of existing prototype devices[6] poses significant challenges for future practical implementation. Researchers are also reviving ancient endothermic cooling techniques – dating back to Asia and Arabia in the 4th century or earlier[7] – based on salt and ice or water dissolution, which do not require moving parts. These techniques, however, are traditionally non-reversible, limiting their practical use.

Writing in *Nature Energy*, Si Wu, Ruzhu Wang, Tingxian Li, and colleagues[8] report an endothermic approach based on reversible dissolution-based cyclic cooling. They incorporate a sorption process into the cycle to drive regeneration of a potassium thiocyanate (KSCN) salt endothermically dissolved in water. This approach simultaneously enables heat recovery through the exothermic adsorption of the distilled gas and opens further possibilities for the exploitation of the technology.

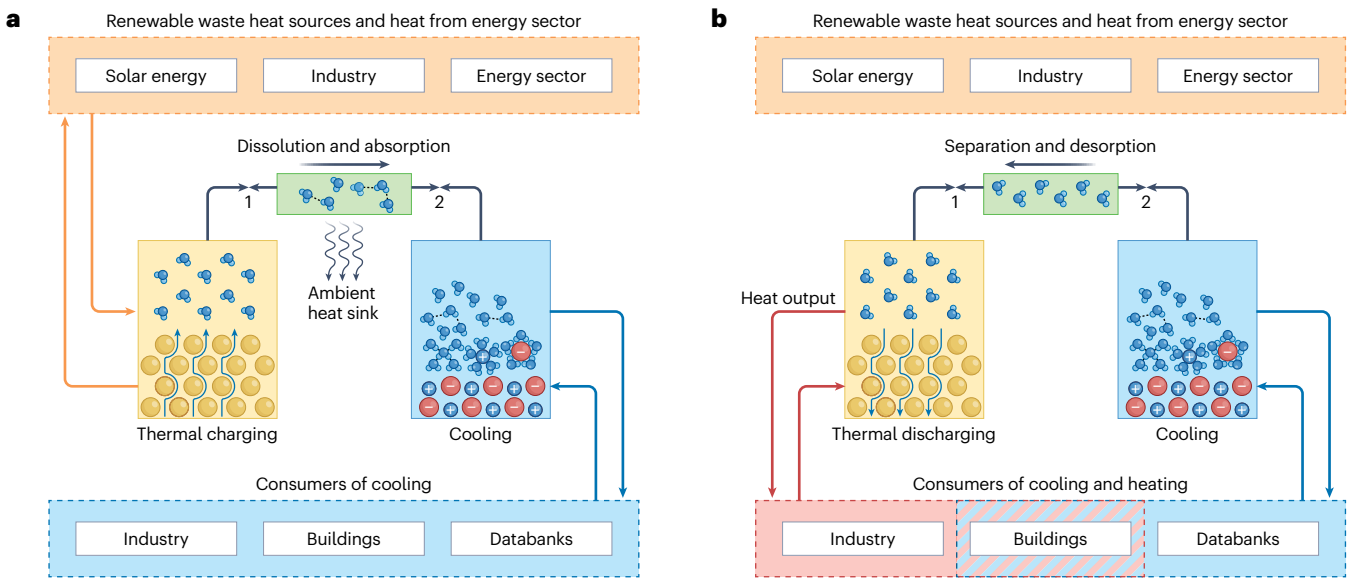

Fig. 1 | Illustration of the sorption-driven dissolution cycle and its potential applications. The yellow, green, and blue squares represent the heat storage tank, condenser, and refrigeration tank, respectively. The yellow circles represent the KSCN sorbent, the hydrated salt is shown with blue and red circles. Water molecules are represented by blue circles, and hydrogen bonding between them is indicated with dashed lines. **a**, Schematics of the cycle during simultaneous charging and cooling. The orange box at the top shows renewable and waste sources: solar energy, industrial waste heat, and heat produced in the energy sector (that is, power plants and district heating, and electrode boilers). The blue box at the bottom shows different users of cooling: industry, the building sector, and databanks. **b**, Schematic of the cycle during simultaneous discharging, heating and cooling. The box at the bottom shows users of heat (red), including industry and the building sector, and users of cooling (blue), including the building sector, and databanks. Figure adapted from ref. 8, Springer Nature Ltd.

The system works by using waste heat or renewable energy to desorb water vapour from the KSCN sorbent, which then condenses into water and releases heat to the environment (Fig. 1a). The thermal energy is thus stored in the sorption tank. Opening the valve between the condenser and refrigeration tank allows condensed water to mix with the salt, an endothermic reaction that causes rapid cooling, dropping the temperature. After cooling, the valves close to separate the sorbent and the solution. Reopening the valves restarts heat generation via sorption for reuse of stored heat (Fig. 1b).

A key criterion in salt selection is its potential energy density and adiabatic temperature. Energy density is directly related to cost per unit power and helps minimize the device's carbon footprint per unit mass. The adiabatic temperature change determines cooling performance. Other important factors include cyclic stability and environmental impact. With respect to the latter, KSCN and its aqueous solution have zero global warming potential, ozone depletion potential, and do not form trifluoroacetic acid — metrics that indicate minimal contribution to greenhouse gas emissions. Compared with other dissolution refrigeration salts such as NH_4NO_2, NH_4CN, KCN, and NH_4I, KSCN is of relatively low toxicity.

The methodology developed by Wu, Wang, Li, and co-workers substantially outperforms both the power density and the adiabatic temperature change achieved by other reported dissolution refrigeration methods reported in the literature, such as the approach of Drew Lilley and Ravi Prasher[9], who mixed sodium iodide salt with ethylene carbonate to reduce the melting point and induce cooling, and then employed electrodialysis to reverse the process. Moreover, the method of Wu, Wang, Li, and co-workers is complementary to the work of Seonggon Kim et al.[10], who used ammonium nitrate or potassium nitrate dissolved in water. While the dissolution approach of mixing salts with water is similar, the two studies differ in the method used to reverse the process. Wu, Wang, Li, and team employed sorption, whereas Kim et al. utilized electrodialysis. However, the work of Wu, Wang, Li, and co-workers surpasses that approach in different ways: by introducing non-explosive, environmentally benign, and low-toxicity salts; by improving the second-law efficiency (about 30% versus 20% of Kim et al.); and by enabling the use of renewable heat sources and abundant waste heat.

The potential of the sorption-driven dissolution concept could in principle go beyond refrigeration. In their study, Wu, Wang, Li, and co-workers also describe several options for implementing the sorption-driven dissolution refrigeration cycle for continuous cooling or heating, simultaneous cooling and heating, daily or seasonal thermal storage, and combinations thereof (Fig. 1). These configurations can be adapted to diverse thermal-management scenarios in heat-intensive processes, buildings, various industries, and district energy systems. The inherent modularity and configurational flexibility further position sorption-driven dissolution concepts as promising components of Carnot-battery systems that support energy-sector coupling — a distinct advantage over alternative cooling technologies.

Several key challenges remain for future development. Mature vapour compression technology achieves power densities ranging from 0.5 kW kg^{-1} for small devices to 25 kW kg^{-1} of refrigerant for large-scale units[4]. By comparison, the average power density of 0.17 kW kg^{-1} achieved by Wu, Wang, Li, and team during cooling (dissolution stage) is modest when compared to some solid-state caloric technologies. However, it can be substantially improved through enhanced and faster cooling and other process optimizations. Notably, the study demonstrates superior adiabatic temperature changes of 37 K — something that would require enormous magnetic fields, electric fields, or strain in solid-state caloric systems. Future progress also hinges on enhancing heat and mass transfer. The research team emphasizes that advanced sorbents like metal–organic frameworks — with low desorption temperatures and high capacities — are essential for low-grade heat utilization and energy density improvement. Methods to accelerate dissolution/crystallization, along with solutions for vacuum maintenance, corrosion mitigation, and long-term cyclic stability improvement, must also be developed to strengthen the technological viability of sorption-driven dissolution technologies.

Despite these challenges, the work of Si Wu, Ruzhu Wang, Tingxian Li, and colleagues offers a viable alternative to traditional sorption or vapour compression, complementing solid-state caloric technologies and other alternatives that typically require mechanical work or electricity. Sorption-driven dissolution refrigeration through the effective utilization of waste or renewable heat could constitute an important pillar for sustainable cooling and heating within future energy infrastructure.

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Competing interests

The author declares no competing interests.